



Chapter 8: Search Trees



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Outline

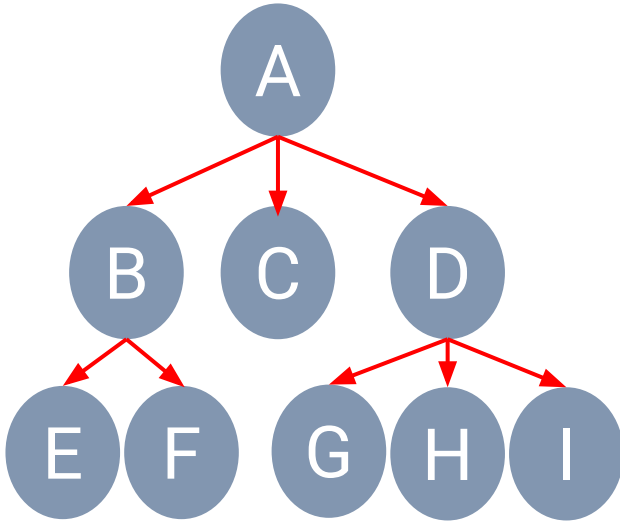


Binary Search Trees:

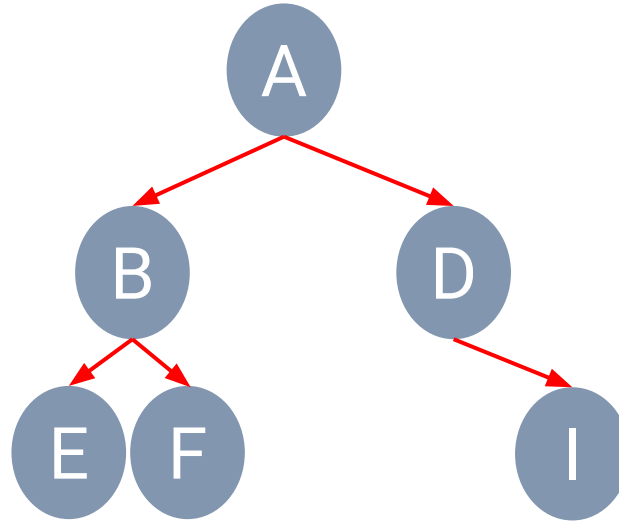
- Definition, properties and methods (search, add, delete)
- Algorithms and Operation examples



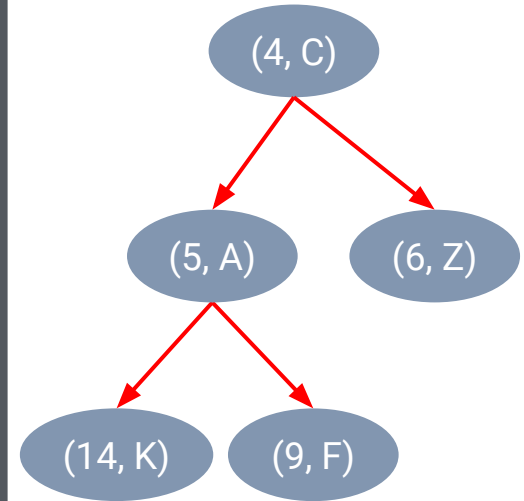
Types of Trees (Revisited)



General Tree

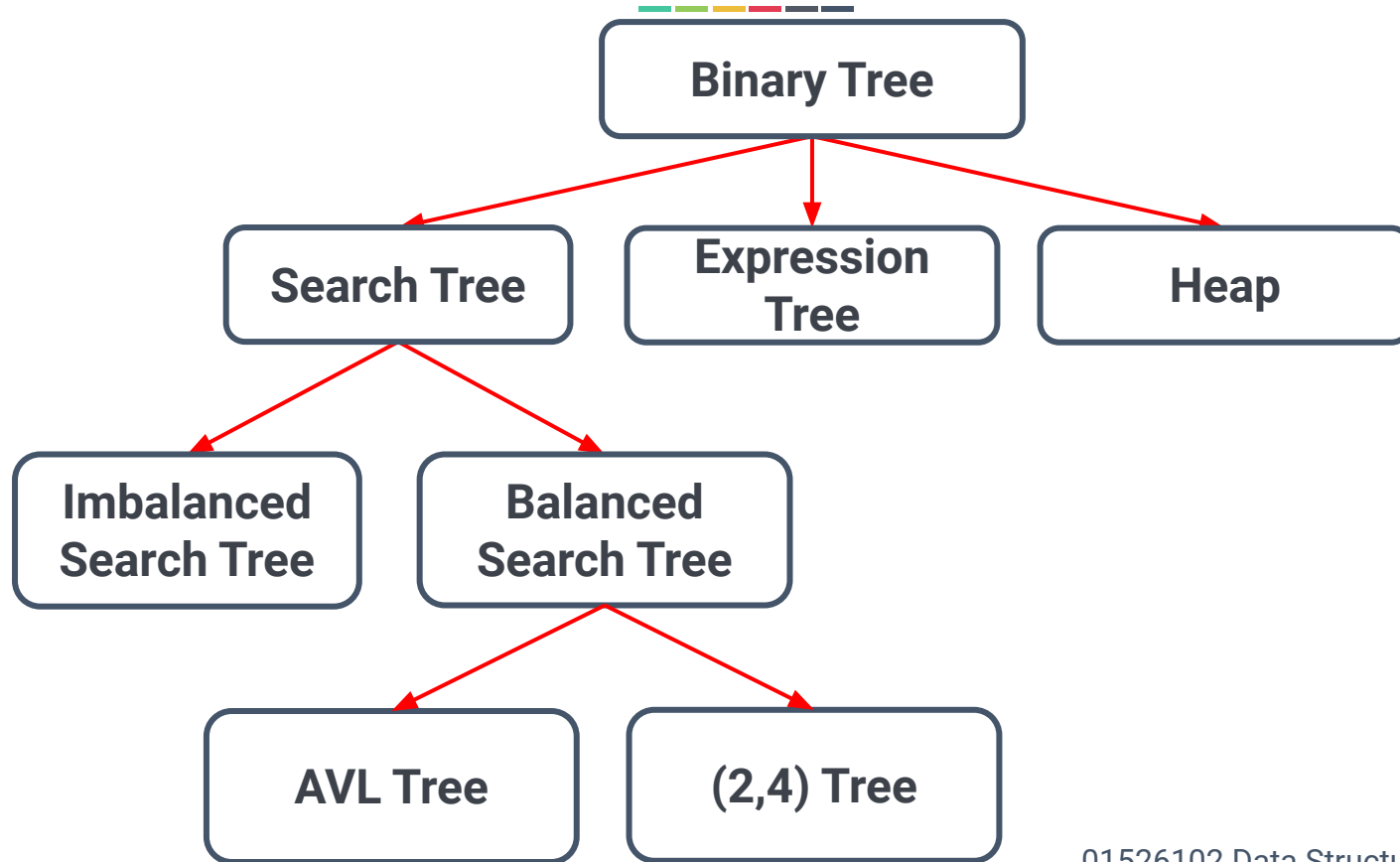


Binary Tree



Binary Heap

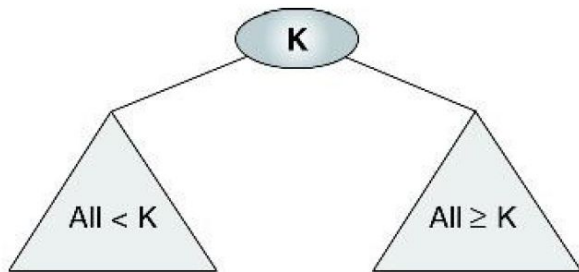
Types of Binary Trees



What is a Binary Search Tree?



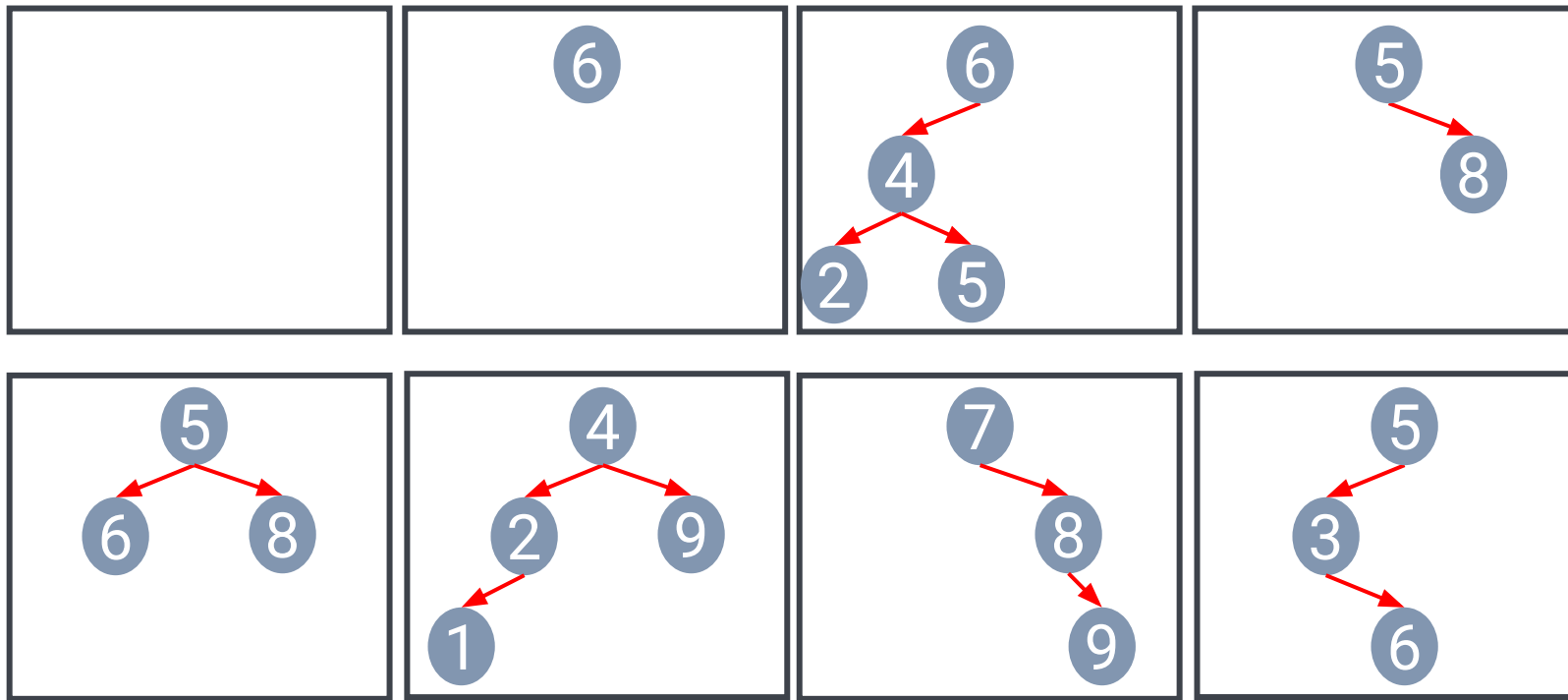
- A **binary search tree (BST)** is a binary tree that stores an ordered sequence of elements or pairs of keys and values and has the following properties [1]:



- All keys/elements in the **left subtree** are **less than** their **root**.
- All keys/items in the **right subtree** are **greater than or equal to** their **root**.
- Each subtree itself is a binary search tree.



Valid/Invalid Binary Search Trees?

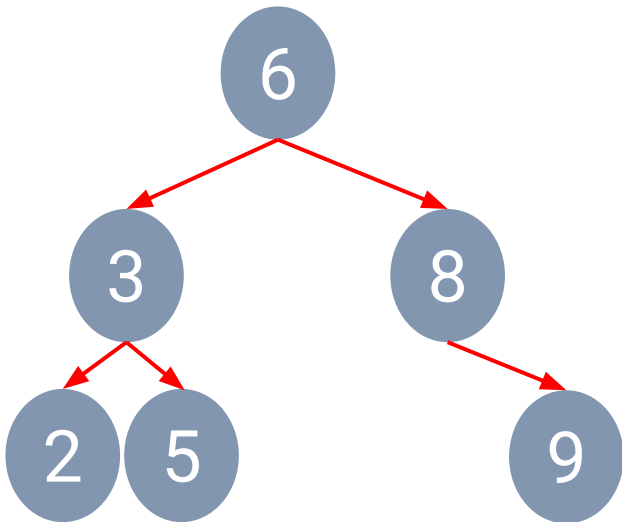




Binary Search Tree



- A **binary search tree (BST)** applies the inorder traversal algorithm to insert keys and navigate the tree.
 - Produces a sorted keys in linear time.





BST Node Implementation



```
class BST_Node:
```

```
    def __init__(self, key, val, left=None, right=None, parent=None):
```

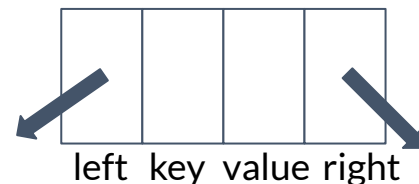
```
        self._key = key
```

```
        self._value = val
```

```
        self._leftChild = left
```

```
        self._rightChild = right
```

```
        self._parent = parent
```



```
def ....
```




BST Implementation



```
class BinarySearchTree:
```

```
    def __init__(self):
```

```
        self._root = None
```

```
        self._size = 0
```

```
    def ....
```

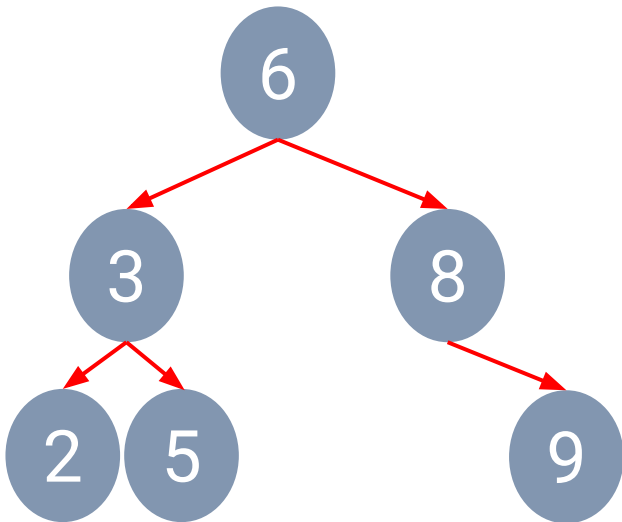
```
    def ....
```



Search in Binary Search Tree



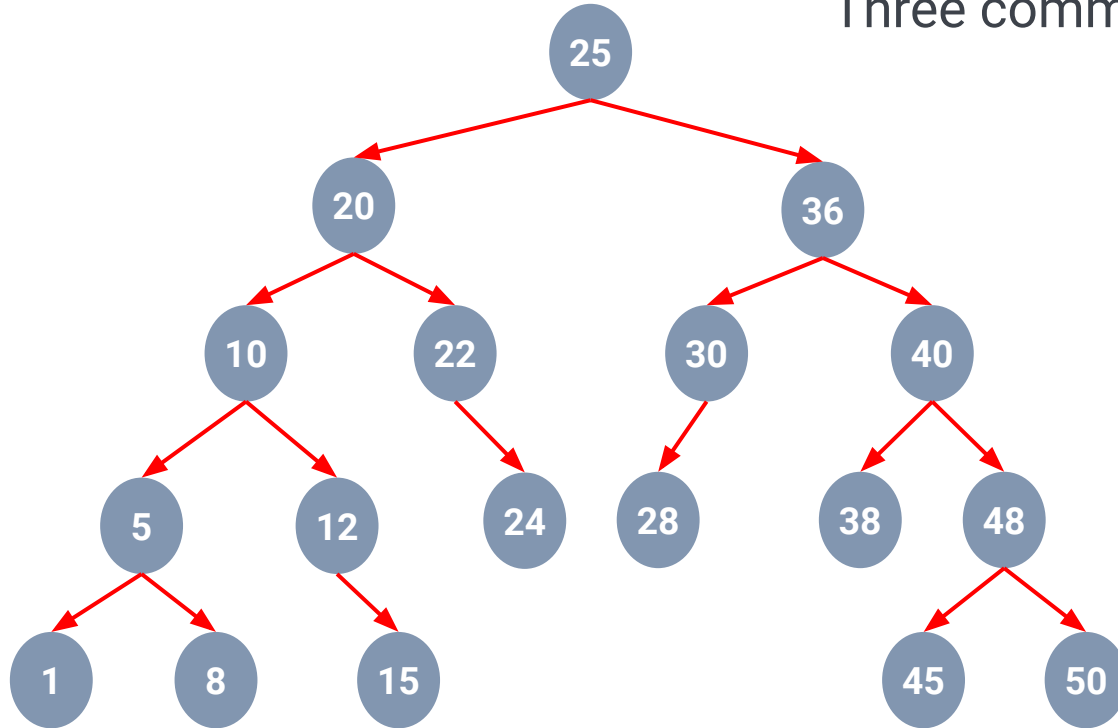
- A **binary search tree (BST)** can be used to find whether a given key is stored in a tree by starting at the root.
 - For each position p , the searched *key* are compared with the key stored at position p , which is denoted as $p.key()$.



Searching in BST



Three common cases of searching in BST:





Binary Search Tree Algorithm





BST Construction and Insertion





BST Construction and Insertion



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BST Insertion Exercise 1



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BST Construction and Insertion Algorithm



Algorithm treeInsert(key, value):

if rootNode is not empty **then**

 treeInsert(key, value, rootNode) # Root node exists

else:

 create a new tree and put key & value as the root #Root node not exists

self._size = self._size + 1



BST Construction and Insertion Algorithm



Algorithm _treeInsert(key, value, currentNode):

if key < currentNode.key **then**

if # Recur on left subtree

else

else:

if # Recur on right subtree

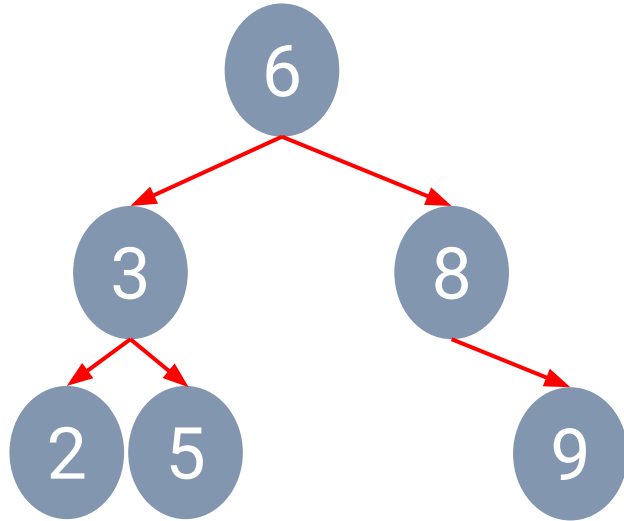
else



BST Deletion



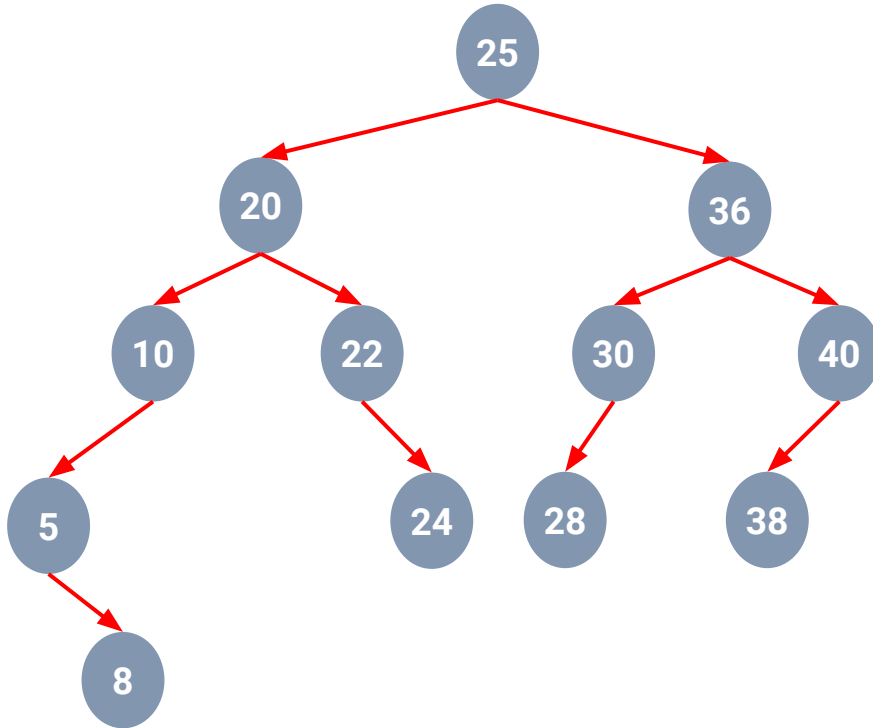
- The deletion of a key from a BST can be performed on any node.
- Once a target node is found, three cases to consider:



BST Deletion Case #1



#1: The node has **zero** child.





BST Deletion Case #1



#1: The node has **zero** child.



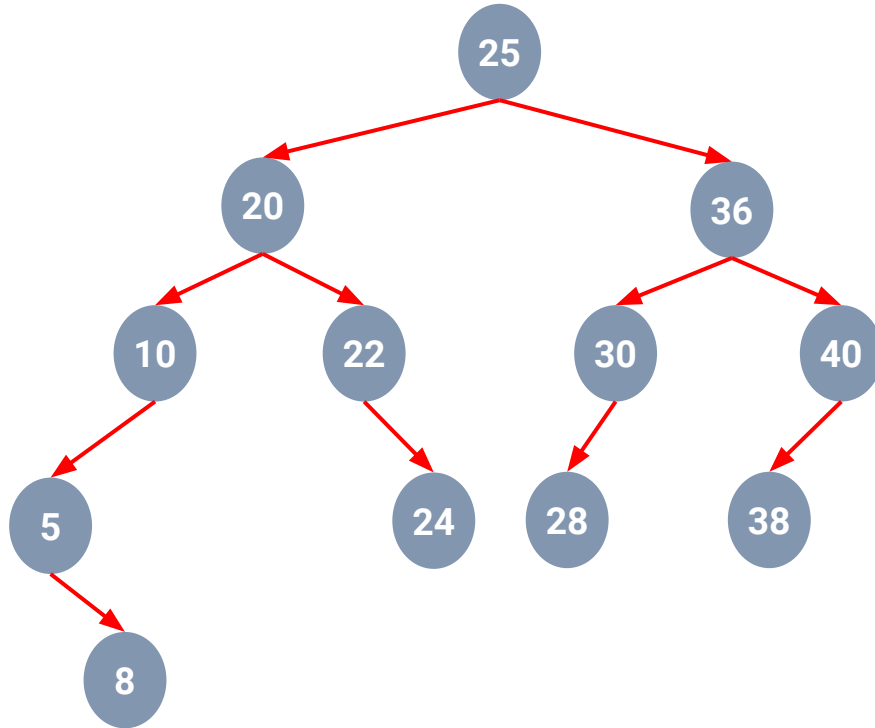
BST Deletion Case #2



If a node has only a **single** child, the child could be promoted and replace its parent.

- 6 Sub-cases to consider:
 - A node has a left child
 - A node itself is a left child.
 - A node itself is a right child.
 - A node itself is the root node, no parent.

BST Deletion Case #2



#2: The node has **one** child.



○ A node has a left child

■ A node itself is a left child.



BST Deletion Case #2



#2: The node has **one** child.



BST Deletion Case #3



- If a node has **two** children, one of the child could not be simply promoted and replace its parent since the other child would be left out of the tree.



BST Deletion Case #3

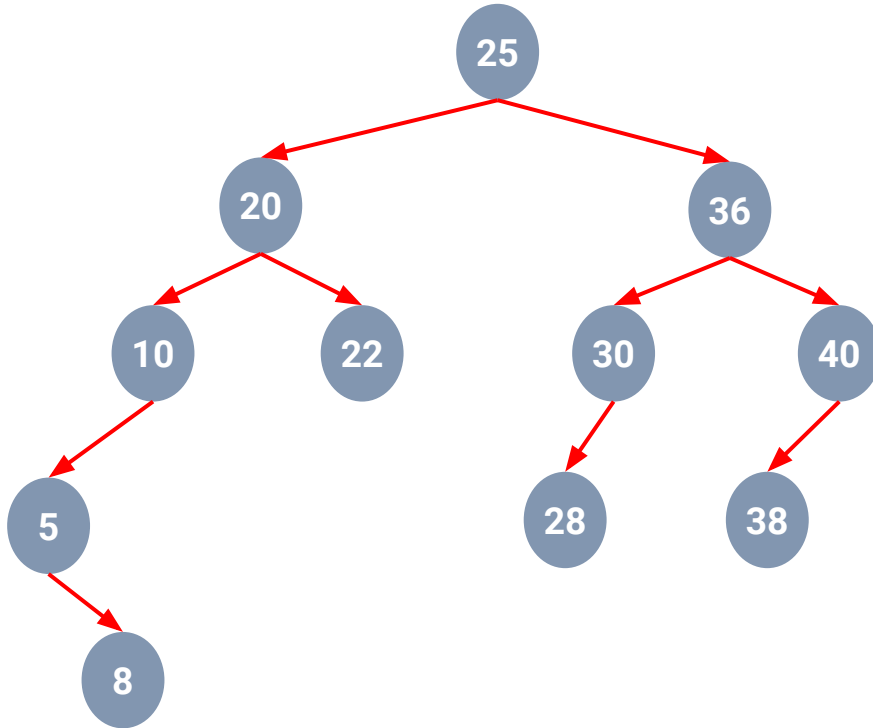


- To preserve a binary tree property, need to search the tree for a **successor** node, which is the next largest key after the deleted node:
 - If the successor has one child, it can be removed using case #1 or #2.
 - Then the successor node replaces the deleted node.
- The successor node could be either:
 - The minimum key node in the right subtree. Or
 - The maximum key node in the left subtree.

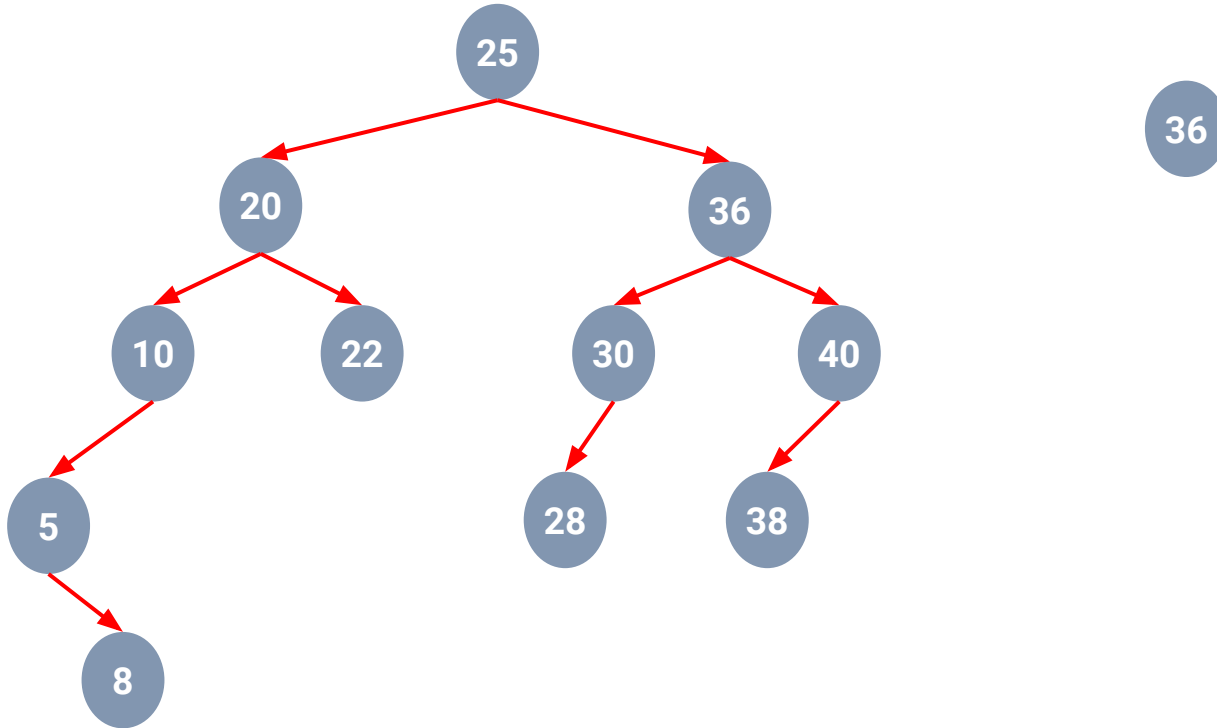
BST Deletion Case #3



#3: The node has **two** child.



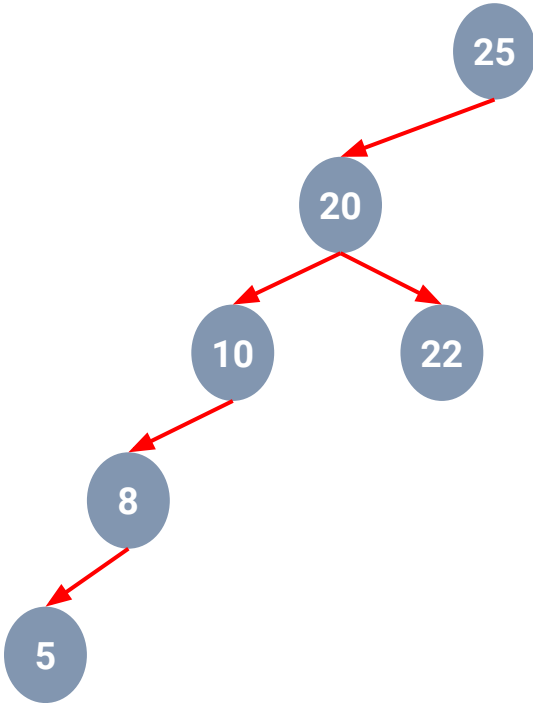
BST Deletion Exercise#1



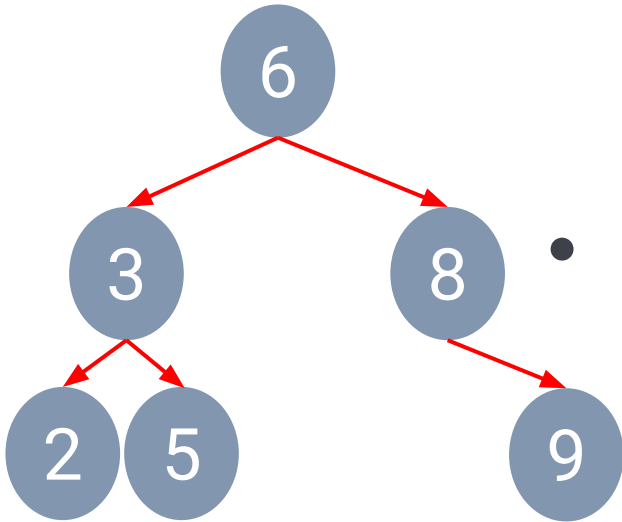
Binary Search Tree



- Running time of inserting node is also proportional to the **height of tree** (i.e. $\log_2 n$ or n) == $O(h)$.
- A **balanced search tree** has the same number of nodes in both left and right subtree.
 - Worst-case performance is $O(\log_2 n)$.
- Inserting keys in sorted order would construct an **imbalanced tree**.
 - Provides poor performance of $O(n)$.



Balanced Binary Search Tree



- A **balanced binary search tree (BST)** maintains the balance through a rotation operation which consequently provides a better performance.
- Several types of binary tree that automatically ensure balance
 - AVL tree
 - Splay tree
 - Red-black tree